Which prescribers should 60 reps call on?

- Branded DOAC, fixed 60-rep field force (~$15M a year fully loaded), flat share.
- 1,380,665 Medicare Part D prescribers, 267,171 of whom write the class; 2022-2024.
- Every number in this deck is generated from the pipeline output.

EXECUTIVE SUMMARY

Re-target on the hybrid ranking (opportunity + volume); hold headcount; redraw the map

- Same 60 reps, same cost: the recommended list captured 2.2% of 2024 share gains vs 1.8% for today's volume list (1.19x), keeping 91% of its absolute growth.
- Flagged prescribers outgrew propensity-matched peers by 4.9 branded fills [3.8, 6.0] in a year the model never saw.
- Headcount: hold at 60. Public data show no measurable call response; the 60th rep pays only if a year of calls captures >= 1.72% of remaining opportunity. A 20% hold-out scored on brand-share points measures that with 90% power in two quarters.
- Territories: imbalance 106x -> 3.6x, contiguity 100%, vs plain k-means at 523x and 98%.

Evidence classes: BACK-TESTED (validated on 2024), ARITHMETIC (no behavioural assumption), SCENARIO (rests on the fitted response curve).

SITUATION & COMPLICATION

Volume ranking cannot see headroom

- The default rule ranks prescribers by how much of the class they write.
- A prescriber already at 90% brand share has nothing left to win.
- A mid-volume prescriber at low share, with an elderly high-risk panel, has a lot.
- The two rankings disagree by 2+ deciles for 52.0% of in-market prescribers.

APPROACH

Model what each prescriber could reach, then subtract

- Quantile-regression frontier (tau 0.80): what a STRONG comparable writer reaches.
- opportunity = potential class volume x peer-achievable share - current brand fills
- In-sample coverage 0.800 vs tau 0.80; linear-frontier rank agreement 0.867.
- Importance: practice size & patient mix 76%; specialty 18%; local market (geography + PLACES prevalence) 6%
- (It is mostly a practice-size and patient-mix model; geography adds little.)

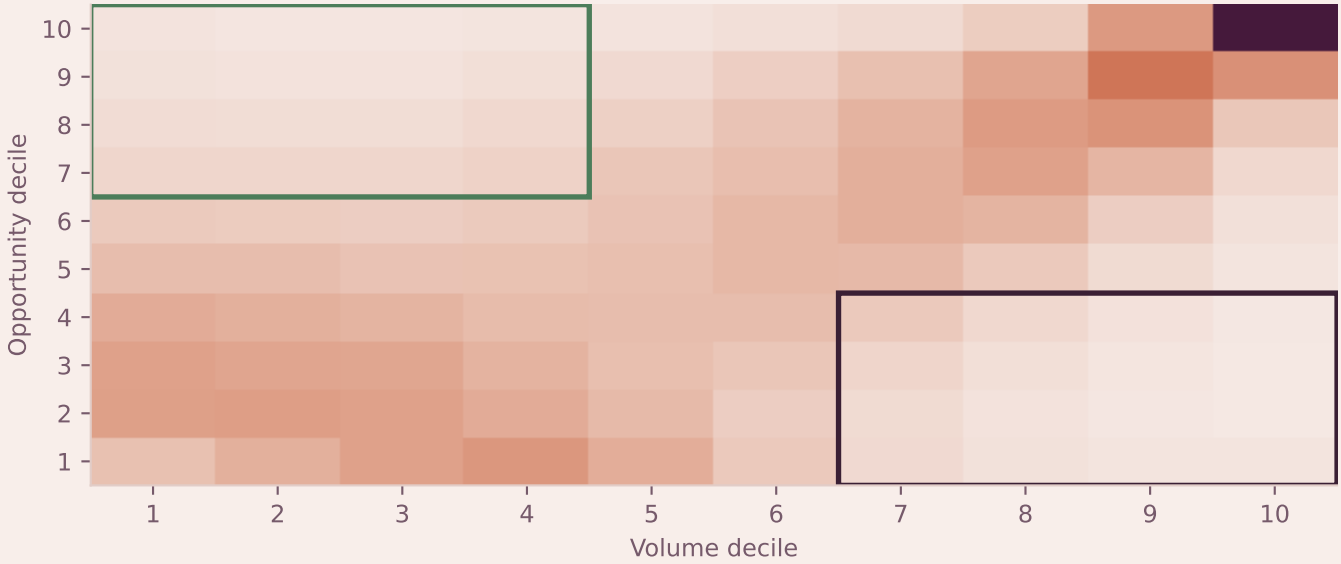
DATA

Streamed ~39 GB of public CMS files down to ~1.3 GB

- 13.7% of all Part D claim volume is suppressed (rows under 11 claims are removed, not blanked).
- Hidden share by prescriber size: <100 claims 68.7%, 100-500 43.3%, 500-2,000 21.6%, 2,000+ 8.6%
- Recovered by reconciling provider totals against drug rows, capped at what the
- unobserved class drugs can physically hold; share is measured on observed volume.
- Promotion is attributed by PRODUCT name, not just by company.

EXHIBIT E1 / ARITHMETIC

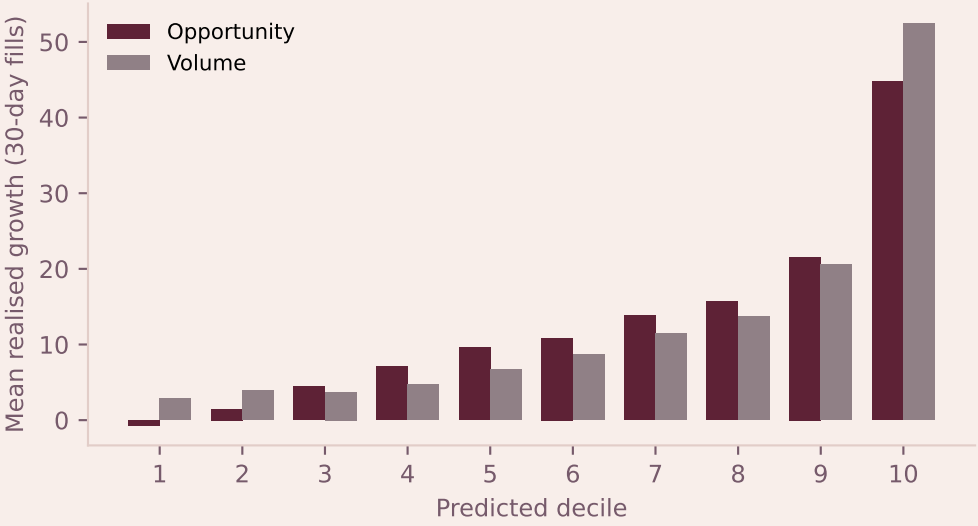
Volume and opportunity disagree for 52.0% of prescribers



Source: CMS Medicare Part D & Open Payments, 2022-2024.

EXHIBIT E2 / BACK-TESTED

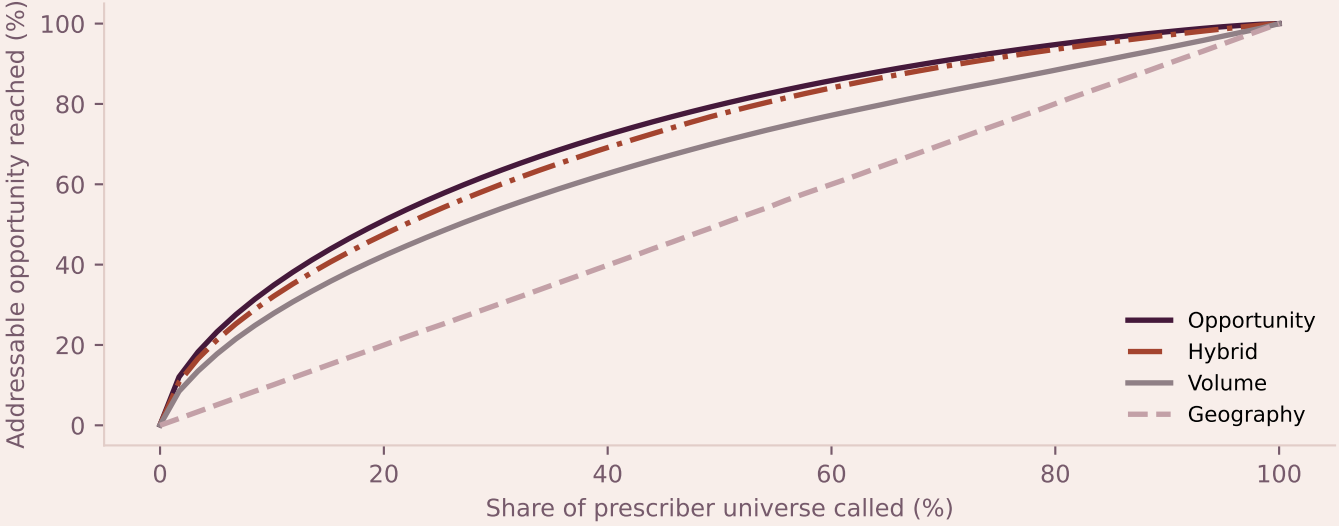
Held-out year: growth rises with the decile (prescriber-level rho 0.22 [0.2



Source: CMS Medicare Part D & Open Payments, 2022-2024. Fit on 2023, frozen, scored against 2024. Volume wins absolute growth at the top.

EXHIBIT E3 / ARITHMETIC

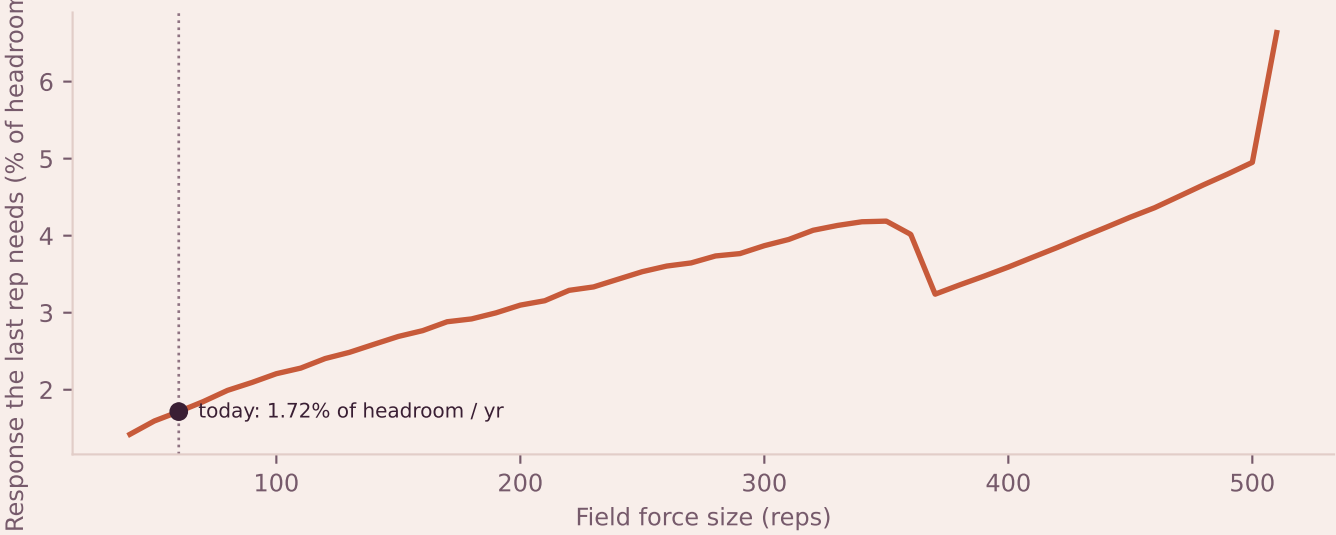
Same budget, 18.0% reached vs 13.3%



Source: CMS Medicare Part D & Open Payments, 2022-2024. No behavioural assumption: coverage, not outcome.

EXHIBIT E4 / SCENARIO

What each force size would need calling to deliver

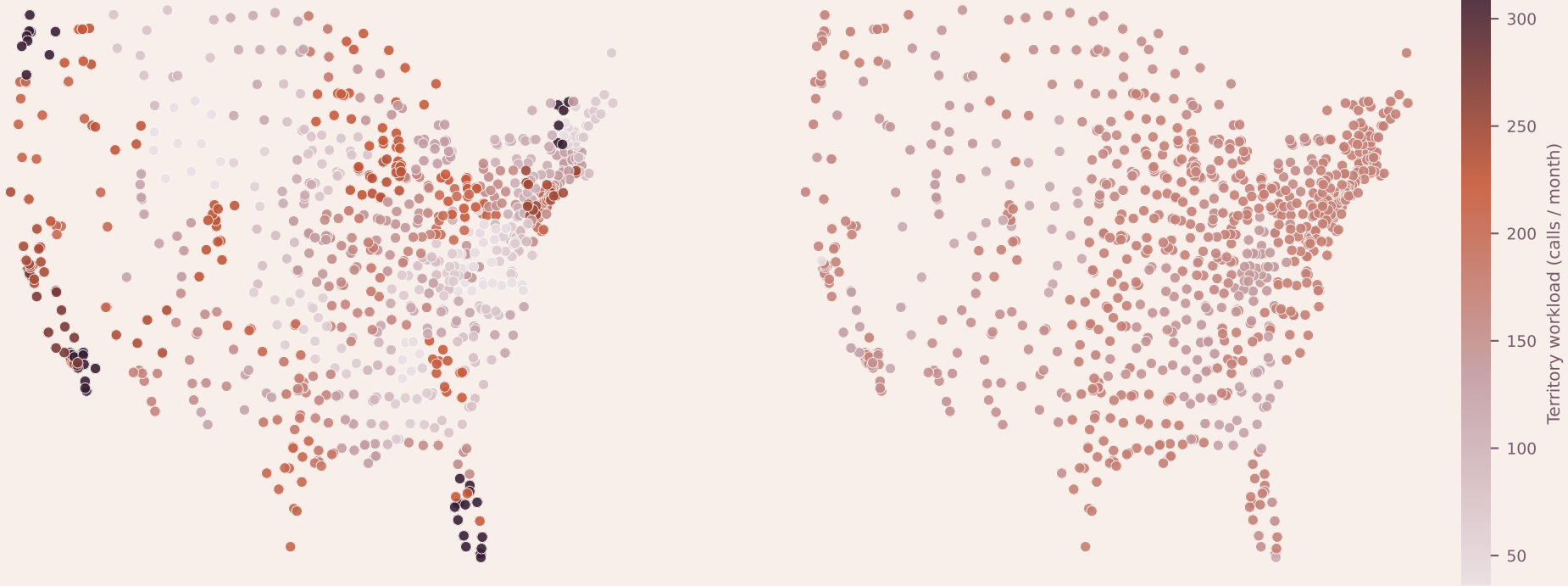


Source: CMS Medicare Part D & Open Payments, 2022-2024. Public data show no measurable response (fit not identified). Half-saturation assumed at the c

Territory alignment: imbalance 106x -> 3.6x | contiguity 20% -> 100% | travel -70%

Before - alphabetical by state

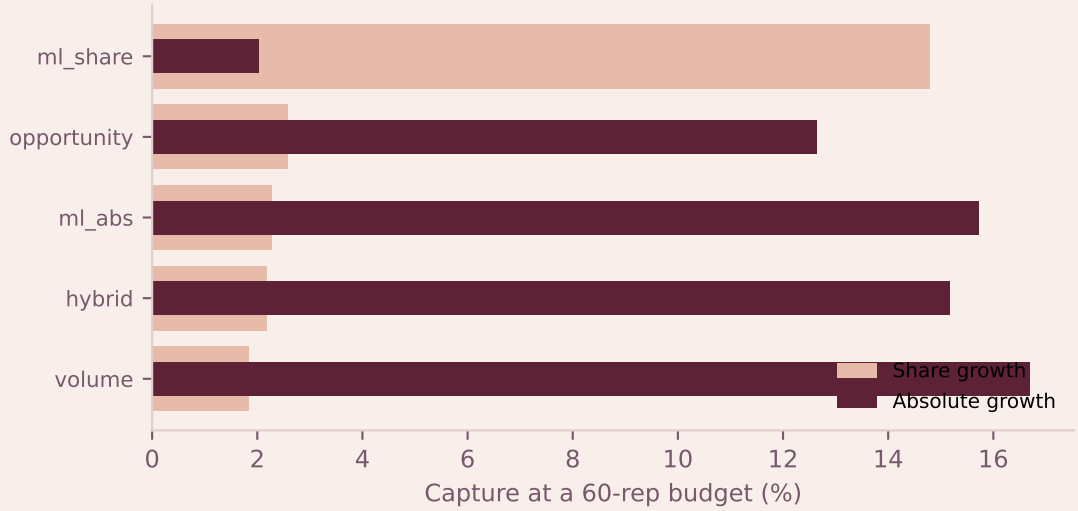
After - capacity-constrained, contiguous



Source: CMS Medicare Part D & Open Payments, 2022-2024. Each dot is a ZIP3, coloured by its territory's total workload on a shared scale. Contiguous US shown; all units included in the solve.

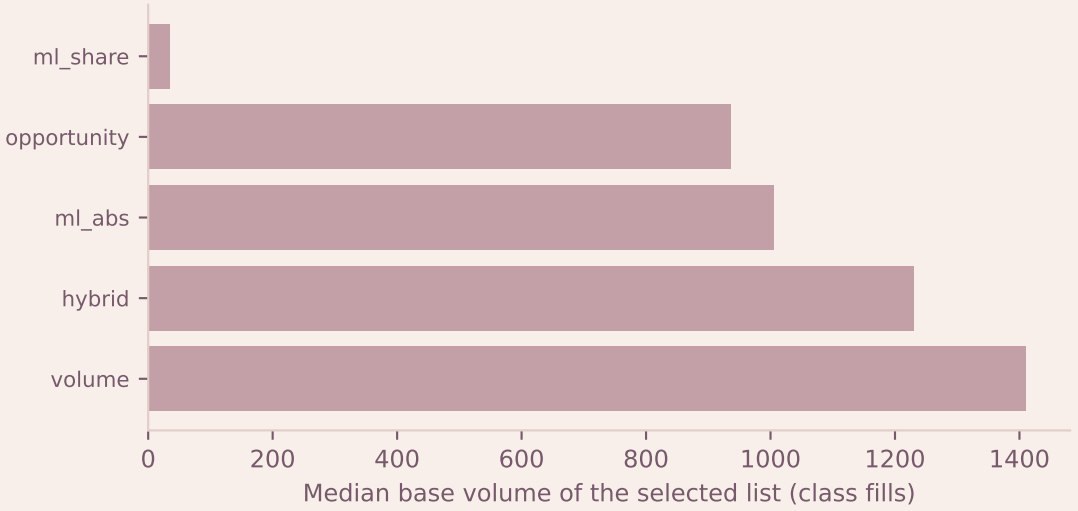
CHAMPION VS CHALLENGER

The ML model wins the stated metric



THE DIAGNOSTIC

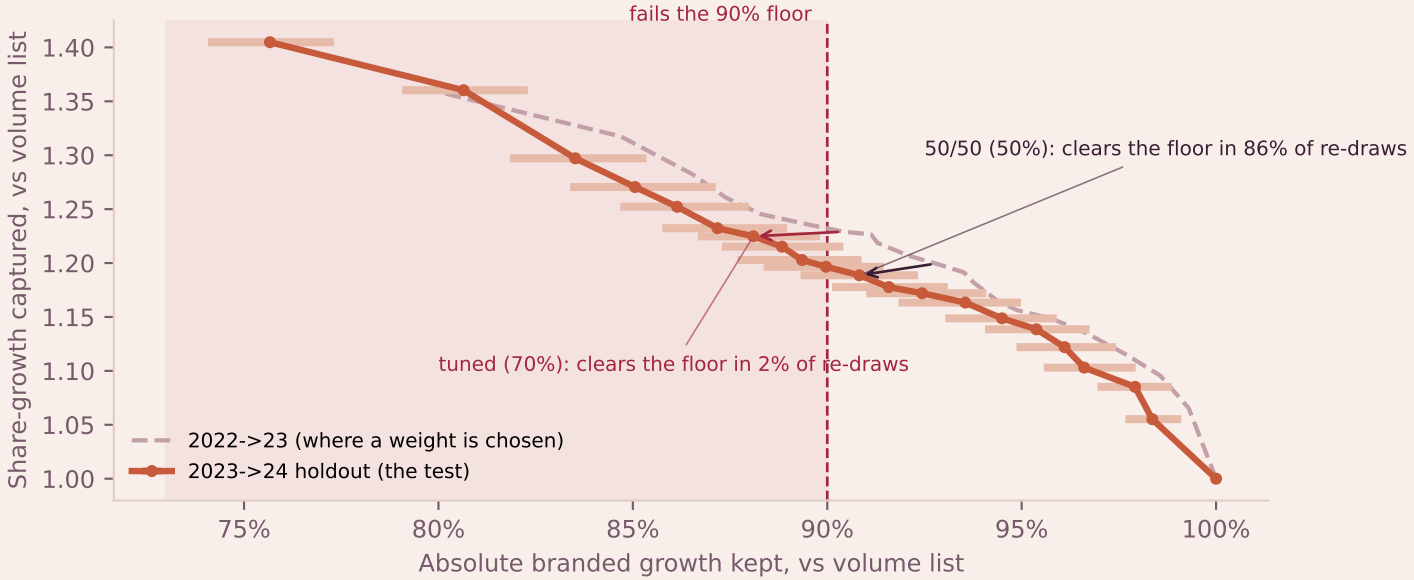
...by selecting micro-prescribers



Source: CMS Medicare Part D & Open Payments, 2022-2024. Share growth on a tiny denominator is arithmetic, not persuadability.

EXHIBIT E7 / BACK-TESTED

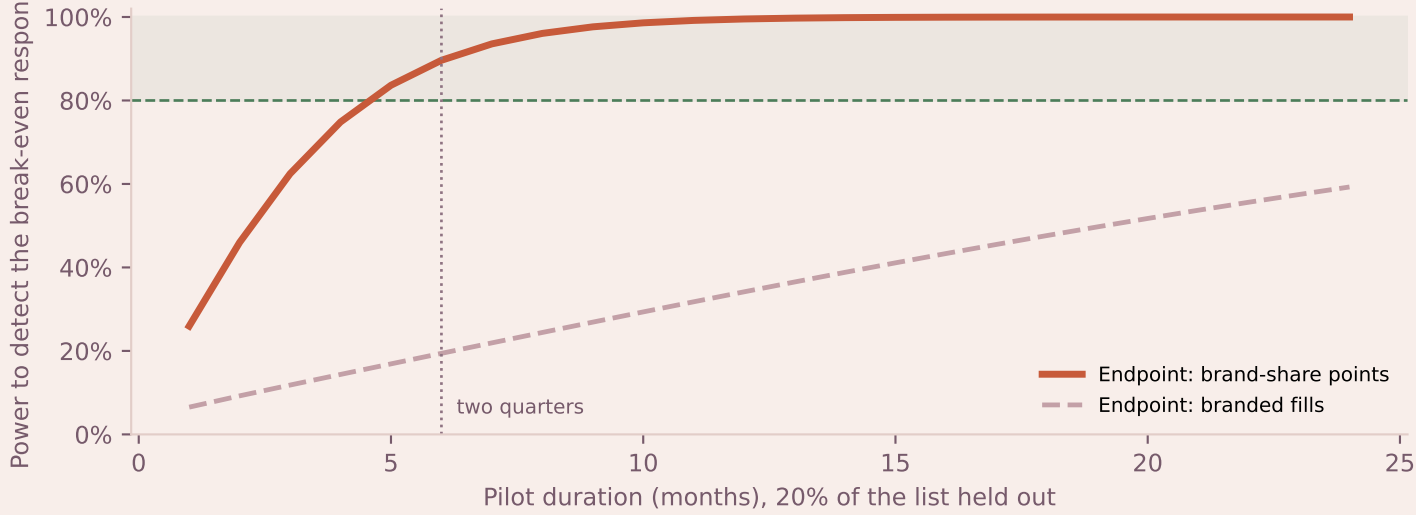
A weight tuned on last year falls through the floor; 50/50 holds



Source: CMS Medicare Part D & Open Payments, 2022-2024. Shaded bars: 95% bootstrap interval, 200 prescriber re-draws.

EXHIBIT E8 / SCENARIO

Scored on share points, a two-quarter pilot can find the answer



Source: CMS Medicare Part D & Open Payments, 2022-2024. Noise: real 2023->24 change on the listed prescribers, CUPED-adjusted.

FINDING 1 | BACK-TESTED

The ranking predicts where share moves

- Fit on 2023, frozen, scored on 2024. Flagged prescribers vs 39,727 propensity-matched controls: 1.40x the branded growth (worst SMD 0.053).
- Share-growth capture vs volume: 1.25 [1.24, 1.27] (95% bootstrap CI).
- Absolute-growth capture vs volume: 0.82 [0.81, 0.83] -- volume wins this.
- Prescriber-level rank correlation with realised growth: 0.222 [0.218, 0.225].
- (Measures where growth happened, not where a call would have caused it.)

FINDING 2 | BACK-TESTED

Each pure rule wins one metric; the blend is the decision

- Realised 2024 outcomes of each rule's 8,878-prescriber list (what 60 reps serve):
- volume ranking: share 1.8%, absolute 16.7%, net 731,958 fills
- opportunity ranking: share 2.6%, absolute 12.6%, net 619,001 fills
- hybrid ranking (opportunity + volume): share 2.2%, absolute 15.2%, net 715,980 fills
- Pre-declared rule: highest share-growth capture among ['opportunity', 'volume', 'hybrid'] with absolute-growth capture >= 90% of the volume rule's -> hybrid ranking (opportunity + volume).

FINDING 2B | BACK-TESTED

50/50 is robust because it was not tuned

- Weights swept 0-100% opportunity. Chosen on 2022->23, the floor-maximising weight is 70%: on the 2024 holdout it keeps 88.1% of volume's growth and clears the floor in 2% of re-draws.
- Equal weight keeps 90.8% and wins 1.19x the share gains; it clears the floor in 86% of re-draws.
- The frontier keeps its shape but shifts a couple of points a year: a weight set exactly on the floor is a coin toss the following year.
- If the brand team wants 95% confidence of keeping the base: 45% weight, at a small cost in share gain.
- (Re-read the floor at every annual data refresh.)

FINDING 3 | SCENARIO

Headcount cannot be sized from public data -- hold, and measure

- 60 reps serve 8,870 of 267,171 in-market prescribers (3.3%) at full capacity.
- Brand-named visit events show no dose-response after volume stratification (weighted R^2 -0.10); the curve is not identified. Break-even response for the 60th rep: 1.72% of headroom a year.
- Modelled headroom in front of a rep at 60 reps: opportunity list 18.0%, hybrid 16.4%, volume 13.3%.
- Recommendation: hold 60 reps; split districts into a two-quarter pilot to measure it.
- Modelled value of re-targeting at identical cost: $4,172,658 a year, priced at the break-even response (the response at which today's force exactly pays).
- (That dollar figure is MODELLED; the realised comparison is Finding 2.)

FINDING 4 | ARITHMETIC

Redraw the territories, not the headcount

- Alphabetical-by-state: imbalance 106x, contiguity 20%, travel 235 mi.
- Plain k-means: imbalance 523x, contiguity 98%, travel 49 mi.
- Capacitated + contiguity-repaired: imbalance 3.6x, CV 0.129, contiguity 100%, travel 70 mi.

FINDING 5 | ASSOCIATION

Promotion's association with share cannot be separated from prior trend

- Prescribers first paid in 2023 were ALREADY gaining share faster beforehand: 0.0101 share points vs controls (p 0.000).
- Paid vs unpaid prescribers, 2023->2024 share change: 0.0076 share points (naive).
- Minus their pre-period gap: -0.0025; on matched pairs: 0.0120 [0.0064, 0.0176].
- Parallel trends failed, so this is reported as association only (pre-committed).

The endpoint decides whether the pilot works

- Effect to detect = the break-even response: 1.72% of headroom a year on the listed prescribers.
- Scored on brand-share points, a 20% hold-out has 90% power at six months (100% at twelve).
- Scored on branded fills, the same design has 19%: fills are dominated by practice size, share points are volume-neutral.
- Territory randomisation (30 v 30) reaches 72% at six months; ICC 0.010 costs little.
- Read at month six. Below 0.5x break-even: the force is too big. Above 2x: reopen headcount with a measured number.

RED TEAM

The strongest objections, and what survived them

- 'It rewards big prescribers' -- share growth is volume-neutral, and flagged prescribers grew 1.40x as fast as matched controls of the same size. HOLDS.
- '50/50 is arbitrary' -- swept; the tuned weight fails out of time, 50/50 does not. HOLDS.
- 'You found growth, not persuadability' -- true; no call data. The pilot resolves it. OPEN.
- 'Meals are a poor proxy for calls' -- agreed; hence HOLD headcount, not cut. ACCEPTED.
- 'Medicare-only, 17 months old' -- right lens for older AF patients; refresh yearly. MEDIUM.
- 'Price negotiation changes the maths' -- the 2026 Medicare negotiated price ($231 / 30 days) raises the break-even response from 1.72% by ~1.6x. It strengthens 're-target, do not hire'.

ROADMAP & KILL CRITERIA

Thirty-six weeks from sign-off to a measured decision

- Weeks 0-2: sign-off -- freeze the rule, the 90% floor and the share endpoint.
- Weeks 2-6: load the blended list in CRM, brief reps, randomise the 20% hold-out.
- Weeks 6-14: redraw territories and transition accounts (imbalance target < 4x).
- Weeks 6-32: pilot runs unchanged; month-3 check that noise scales as planned.
- Weeks 32-36: read the share-point effect against break-even; size headcount.
- Kill: effect < 0.5x break-even -> shrink/redeploy; blend < floor at refresh -> move to 45%; call compliance < 80% -> fix adoption first.

LIMITATIONS

What this cannot tell you

- Medicare Part D only; ~17-month lag; suppression hides the smallest prescribers.
- No rep call data: the back-test shows where growth happened, not what caused it.
- The visit proxy (brand-named meals) under-counts calls and reps choose whom to see.
- New-writer acquisition is not modelled; non-writers carry no opportunity score.
- Straight-line distance, ZIP3 units, no rep tenure or org boundaries.

NEXT STEPS

What a brand team would do with this

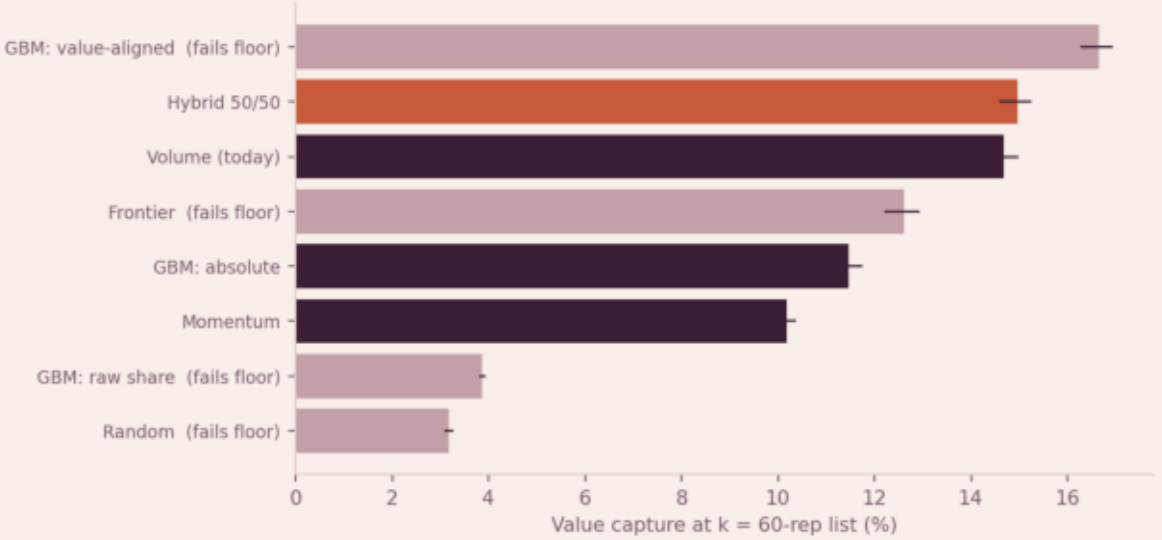
- Pilot: hold out 20% of the new list for two quarters, scored on brand-share change;
- that is the causal test CMS data cannot provide.
- Replace Part D with all-payer claims (IQVIA/Symphony) to remove the Medicare skew.
- Join CRM call logs to replace the meal proxy with observed calls.
- Add drive-time matrices and ZIP5 splits before executing the alignment.

How the models were evaluated, and what they cannot do

- Pre-declared primary metric (value capture at k = 8,878) and guardrail; winner under the guardrail: hybrid.
- Regression-to-the-mean trap found and fixed: corr(share, change) = -0.563; value-aligned target.
- Calibrated 90% intervals: 90.0% in period, 89.3% out of time.
- Double ML: 0.649 pts [0.43, 0.868], robustness value 1.1%.
- Full method and results: docs/TECHNICAL_REPORT.md; model and data cards in docs/.

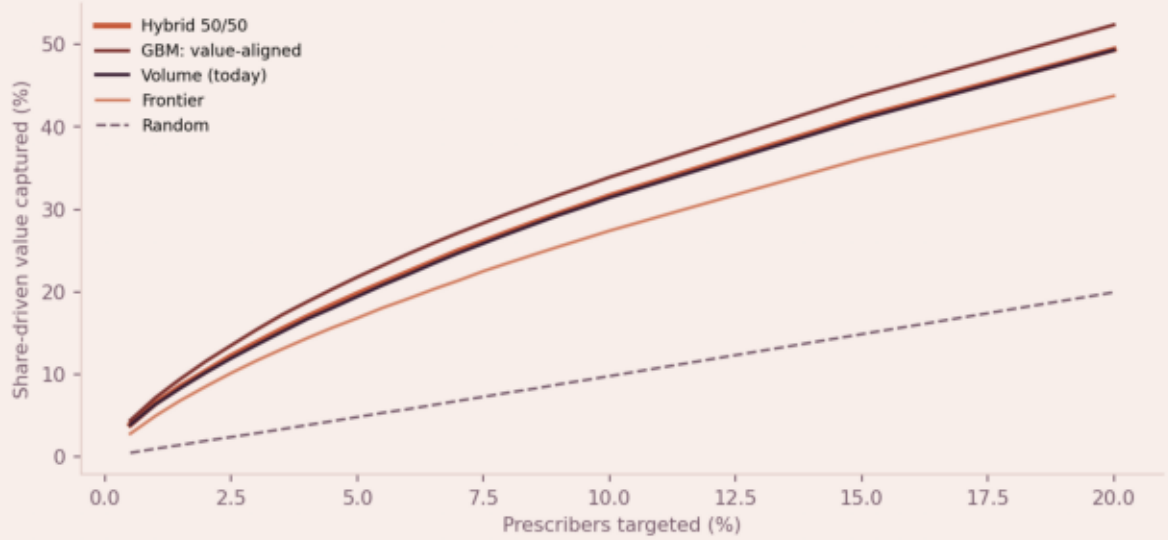
DS1 EVALUATION LADDER

The hybrid is the best rule that keeps the brand's base



DS2 CUMULATIVE GAIN

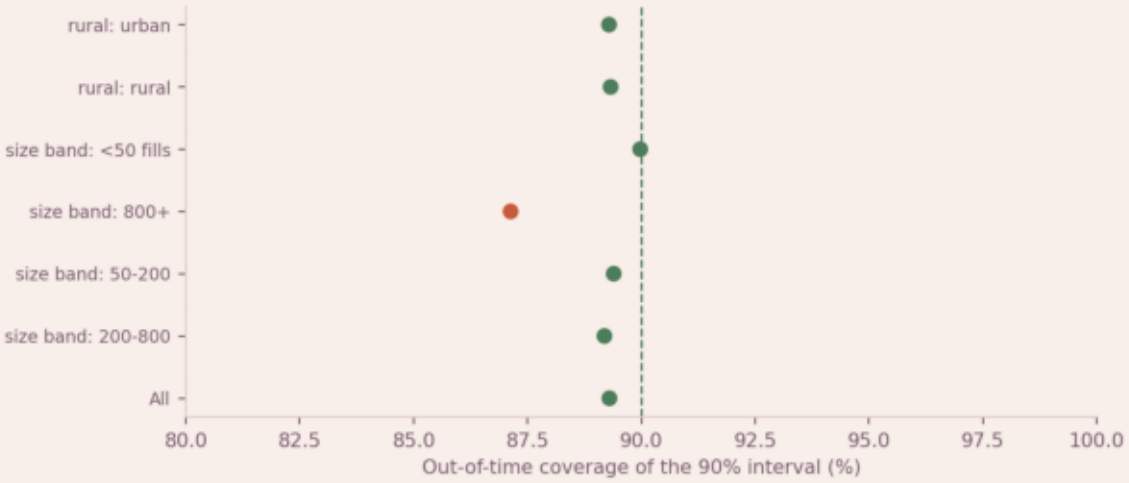
Every serious rule beats random by 3-5x at the 60-rep depth



Source: CMS Medicare Part D & Open Payments, 2022-2024. Data-science appendix; method in docs/TECHNICAL_REPORT.md.

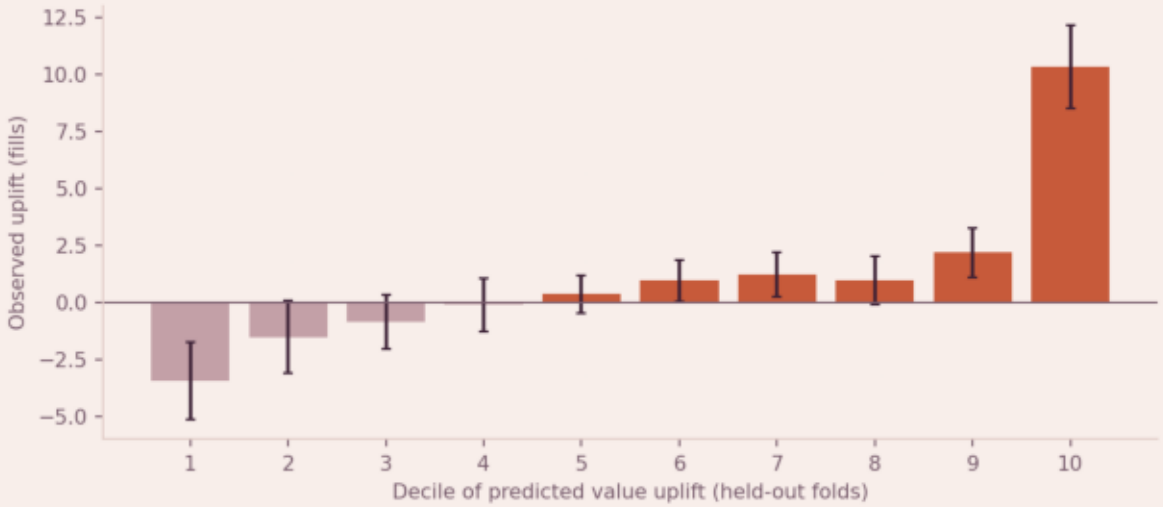
DS3 CONFORMAL COVERAGE

Calibrated intervals hold out of time; drift sits in the largest prescribers



DS4 UPLIFT VALIDATION

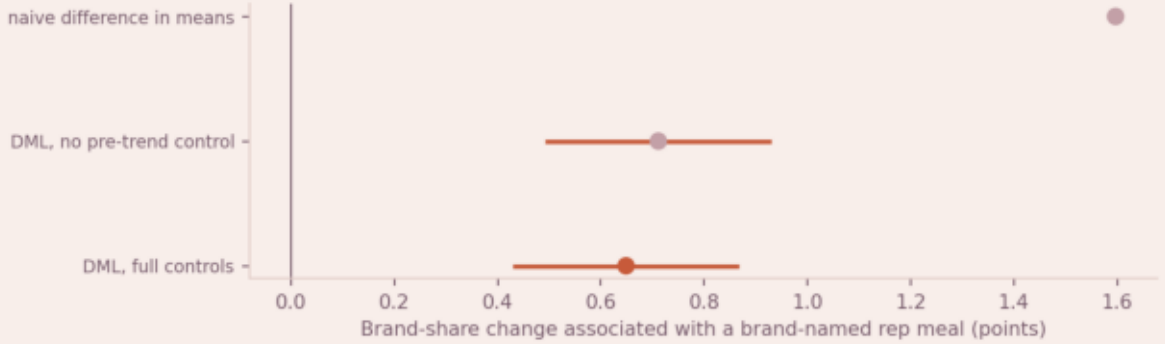
A monotone staircase: the DR-learner ranks who moves when promoted



Source: CMS Medicare Part D & Open Payments, 2022-2024. Data-science appendix; method in docs/TECHNICAL_REPORT.md.

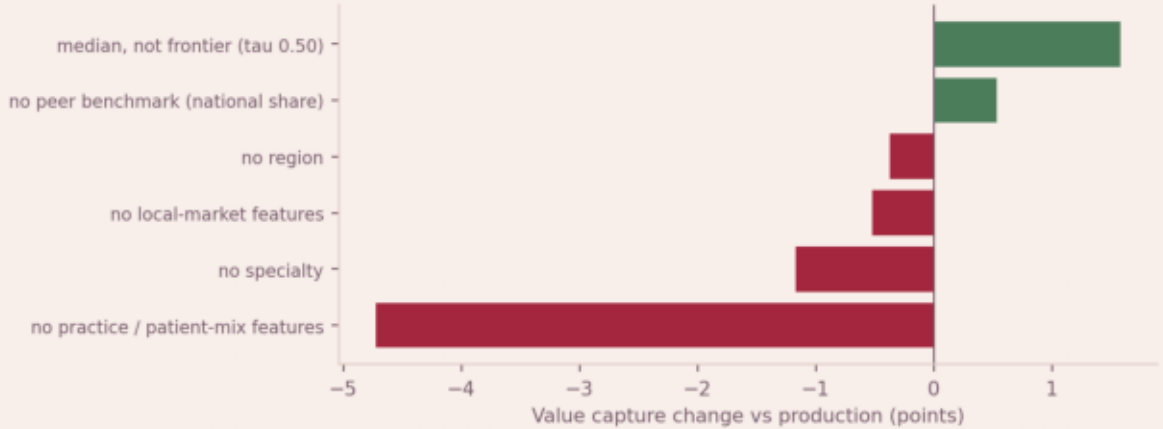
DS5 DOUBLE MACHINE LEARNING

Controls remove 60% of the naive association



DS6 ABLATIONS

Practice size and patient mix carry the model



Source: CMS Medicare Part D & Open Payments, 2022-2024. Data-science appendix; method in docs/TECHNICAL_REPORT.md.